

Modified Sauvé–Kapandji Procedure with Only Extensor Carpi Ulnaris Tenodesis in Advanced-stage Rheumatoid Wrists: A Retrospective Study of Nine Wrists

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Abstract

Background: The modified Sauvé–Kapandji (S–K) procedure with extensor carpi ulnaris tenodesis can treat distal radioulnar joint erosion in patients with rheumatoid arthritis (RA) and reduce proximal ulnar instability. Reports on the functional outcomes after the modified S–K procedure in rheumatoid wrists, especially in severe conditions, are limited. **Objectives:** This study aimed to evaluate the radiographic and functional outcomes after the modified S–K procedure in patients with advanced-stage rheumatoid wrists. **Materials and Methods:** Patients with advanced-stage rheumatoid wrists who underwent the modified S–K procedure between 2012 and 2022 and with ≥ 12 -month follow-up were included. Radiographic examinations and functional assessments were collected through our institution's electronic radiographic, operation, and medical record system. **Results:** Nine wrists from six patients were included; the median age was 57 years (37–79 years), and the median follow-up was 47 months (12–98 months). Radiographic assessments showed no significant differences in the Gilula index ($P = 0.401$) or ulnar gap ($P = 0.678$). Functional outcomes demonstrated enhanced wrist supination ($P = 0.012$) and worse flexion ($P = 0.027$). Although there was a tendency for volarward migration ($P = 0.008$) and radioulnar convergence ($P = 0.021$) of the proximal ulnar stump, improved postoperative quick disabilities of the arm, shoulder, and hand ($P = 0.021$) and modified Mayo wrist ($P = 0.028$) scores were observed. Two patients underwent additional surgery that was not related to the complications of the index procedure. Six wrists of five patients underwent finger extensor tendon reconstruction. Major complications were not observed. **Conclusions:** This study supports the efficacy of the modified S–K procedure in preventing ulnar carpal translation and proximal ulnar pain and improving wrist function in patients with advanced-stage rheumatoid wrists. Although proximal ulnar stump instability may occur, it does not always result in suboptimal clinical outcomes.

Keywords: Extensor tendon reconstruction, proximal ulnar stump, Sauvé–Kapandji procedure, severe rheumatoid wrists, ulnar carpal translation

INTRODUCTION

Individuals with rheumatoid arthritis (RA) frequently exhibit symptoms that affect the following three key wrist articulations: the distal radioulnar joint (DRUJ), radiocarpal joint, and midcarpal joint.^[1] However, DRUJ is especially vulnerable to the detrimental effects of RA and is the most frequently impacted joint.^[1,2] The stability of DRUJ also relies significantly on synovial-lined ligamentous stabilizers, which are also significantly susceptible to the impact of RA.^[2,3] Therefore, joint erosion and ligament injuries are affected by synovial hypertrophy associated with RA, with the production of inflammatory enzymes.^[2]

Osseous DRUJ procedures are indicated in cases of disturbance in wrist articulation, resulting in restricted motion or pain.^[2] The Darrach procedure involves a distal ulnar resection with osteotomy at the proximal margin of the sigmoid notch, which provides pain relief and full rotation, particularly in older patients.^[4,5] However, the Darrach procedure may result in complications, such as proximal ulnar instability, painful proximal ulnar stump, risk of ulnar carpal translation, and loss of grip strength.^[6–8] Alternatively, the Sauvé–Kapandji (S–K)

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procedure involves distal radioulnar arthrodesis and intentional pseudarthrosis of the distal ulna.^[9] Preserving the ulnocarpal joint reduces the risk of ulnar carpal translation and improves long-term outcomes, particularly in younger and more active patients.^[4,10,11] Nevertheless, the impingement and instability of the proximal ulnar stump present some concerns.^[12,13] A modified S–K procedure was introduced to reduce potential complications and stabilize the proximal ulnar stump with the extensor carpi ulnaris (ECU) tendon.^[14] Studies on patients with DRUJ osteoarthritis and RA have demonstrated satisfactory outcomes over long-term follow-up and avoidance of impingement on the proximal ulnar stump via ECU tendon slip.^[15,16]

Most studies focusing on DRUJ procedures have not mentioned the severity of rheumatoid wrists. To the best of our knowledge, limited studies have shown that treating patients with advanced-stage rheumatoid wrists, radiographically defined as Larsen grades III–V,^[17] requires rotating the resected ulnar segment, bridging the gap between the eroded DRUJ.^[18] The results of distal radioulnar arthrodesis without further modifying the DRUJ osseous procedures in patients with severe rheumatoid wrists remain unknown. Furthermore, studies detailing the patient-reported outcome measures of working and daily activities associated with modified S–K procedures are limited, barely mentioning the return-to-work status.^[14–16] Our study aimed to evaluate the radiographic and functional outcomes after the modified S–K procedure in patients with advanced-stage rheumatoid wrists.

MATERIALS AND METHODS

Patients

With the approval of the Institutional Review Board of the E-DA Hospital (EDAH IRB No./ Protocol No: EMRP-113-031, Approval date: 03/20/2024), we performed a retrospective analysis of the surgical records of the International Classification of Diseases-10 code M05.70 between 2012 and 2022 at our institution. This study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki and its amendments. The inclusion criteria were patients with severe rheumatoid wrists who underwent the modified S–K procedure and had ≥ 12 -month follow-up. Patients with other DRUJ operative procedures or a follow-up period < 1 year were excluded from the study. Advanced-stage rheumatoid wrists were defined as Larsen grades III–V based on radiographic findings.^[17] The patients provided informed consent to the photographs taken at the outpatient clinic and data collected from the study.

Operative technique

For patients who presented with DRUJ dysfunction and rheumatoid wrists, the DRUJ operative procedure began with a thorough wrist synovectomy and marginal spur removal. Wrist synovectomy was followed by decortication on the distal ulnar stump and the ulnar side of the radius as an aid to bony union. Subsequently, two steps of osteotomy were performed. In

patients with positive ulnar variance, the first ulnar shortening osteotomy step was performed to correct the ulnar level by using an oscillating saw. Then another ulnar osteotomy of approximately 1 cm was undertaken, followed by ECU tendon transposition. The distal ulnar stump was long enough to fix the two screws for wrist DRUJ fusion. Dorsal subluxation of the distal ulnar was also corrected simultaneously. Care was taken to avoid compressing the radioulnar width while applying the two screws, and the morselized ulnar bone was impacted into the gap of DRUJ. A hole was drilled through the medullary canal to the dorsal cortex of the proximal ulnar stump to prepare for ECU tendon transposition, which was at least 5 mm away from the osteotomy site, to avoid breakage of the drilled hole. To stabilize the proximal ulnar stump, the ECU tendon was split longitudinally from the ulnocarpal level of the musculotendinous junction. Subsequently, the radial half of the released tendon was passed through the drilled hole to complete the stabilization procedure. Finally, finger extensor tendon reconstruction was performed if the extensor tendons were disrupted.

Postoperative volar splinting was performed with immobilization of the affected hand for 2 weeks. Subsequently, active wrist movement was encouraged, and physical therapy was initiated. For patients who underwent tendon procedures, active finger flexion range of motion and passive extension began 2 weeks postoperatively with the protection of the digital extension splint. Wrist and finger motor function, circulation, sensation, and complications were monitored postoperatively and at follow-up.

Radiographic examination and functional assessment

Radiographic parameters included the ulnar distance, ulnar gap, radioulnar distance, dorsopalmar distance, radioulnar width of the wrist, and the Gilula index [Figure 1]. Posteroanterior [Figure 2] and lateral [Figure 3] wrist radiographs were obtained preoperatively and postoperatively at each clinic visit. The radiographic parameters measured the day after the operation and at the latest clinic visit were defined as the early and final postoperative measurements, respectively. The ulnar distance was defined as the remaining length of the distal ulnar shaft.^[19] The ulnar gap was measured from the proximal to the distal ulnar stump.^[13] Radioulnar distance was defined as the interosseous distance between the radius and the distal end of the proximal ulnar stump.^[13] The distance from the dorsal cortices of the radius to the ulna at the level of ulnar resection was labeled the dorsopalmar distance.^[19] The radioulnar width measuring 5 mm proximal to the distal end of the ulnar head was regarded as the radioulnar width of the wrist.^[18] The Gilula index was calculated by dividing the lunate overhang by the lunate's width.^[20]

Preoperative and postoperative range of motion of the wrist and fingers [Figure 4]; grip strength; visual analog scale for pain; Quick Disabilities of the Arm, Shoulder, and Hand (QuickDASH) score; and modified Mayo wrist score (MMWS) were also ascertained through chart review.



Figure 1: Radiographic parameters of the wrist. (a) Posteroanterior view with lines A, B, C, and D representing the radioulnar width of the wrist, ulnar distance, ulnar gap, and radioulnar distance, respectively. (b) Lateral view with line E representing the dorsopalmar distance. (c) The width of the lunate (line F'H') was a line drawn through the most radial point at the middle portion of the lunate to the most ulnar point of the lunate. Point G' was the point of intersection between the line parallel to the long axis of the radius drawn from to the ulnar border of the distal radius and line F'H'. Gilula index = (line G'H'/line F'H')



Figure 2: A 78-year-old man (patient 2) who underwent the modified Sauvé–Kapandji procedure with extensor carpi ulnaris tenodesis for distal radioulnar joint dysfunction. The level of the ulna was corrected postosteotomy. (a) Preoperative posteroanterior view of the wrist. (b) Radiograph taken on the postoperative day 1, showing distal radioulnar arthrodesis and an intentional pseudarthrosis of the distal ulna with ulnar osteotomy and fixation of two headless screws. The radioulnar width of the wrist was maintained. (c) One-year postoperative radiograph of the wrist. (d) Latest radiograph 5 years postoperation

Statistical analysis

SPSS version 25.0 (IBM, Armonk, NY) was used for all statistical analyses. Owing to our small sample size, differences in preoperative and postoperative outcomes were analyzed using the Wilcoxon signed-rank test. Statistical significance was set at $P < 0.05$.

RESULTS

This retrospective study included two males and four females with advanced-stage rheumatoid wrists. Nine wrists of six patients were subjected to the index procedure. Of the nine wrists, six underwent tendon reconstruction. All the procedures were performed by the senior authors. The median age at surgery was 57 years (37–79 years). The median postoperative follow-up period was 47 months (12–98 months).

Patient characteristics are shown in Table 1. Four right wrists and five left wrists were included. Among the nine wrists, three were rated preoperatively as Larsen grade III, four as grade IV, and two as grade V. Among all six hands that underwent tendon reconstruction, five were reconstructed with the palmar longus tendon graft, and one underwent end-to-side tendon repair with the extensor digitorum communis IV.

The results of radiographic examinations are summarized in Tables 2 and 3. A significant difference was observed between the early postoperative and final outcomes in ulnar, radioulnar, and dorsopalmar distances but not in the ulnar gap. The change in radioulnar distance showed a tendency toward radioulnar convergence. The dorsopalmar distance results indicated a tendency for volarward migration of the proximal ulnar stump after surgery. The shrinkage in the ulnar distance was significant, implying bony resorption of the distal ulnar

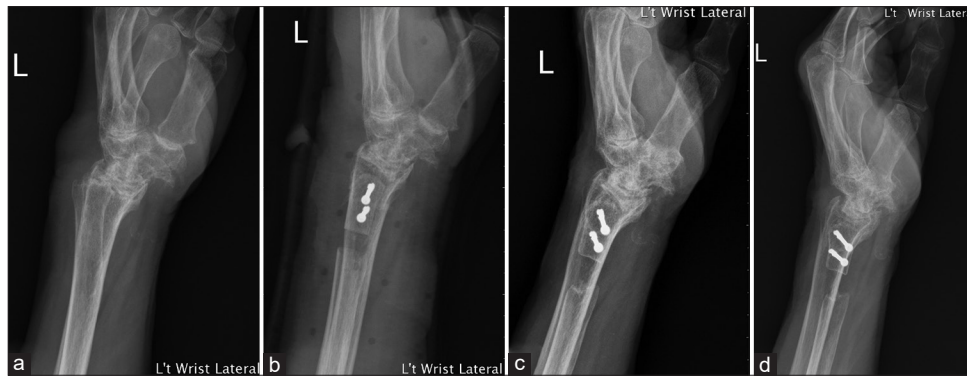


Figure 3: Lateral view of the wrist (the same patient). (a) A preoperative lateral view of the wrist. (b) Radiograph taken on the postoperative day 1, showing distal radioulnar arthrodesis and an intentional pseudarthrosis of the distal ulna with ulnar osteotomy and fixation of two headless screws. Distal radioulnar joint instability was corrected. (c) One-year postoperative radiograph. (d) Latest radiograph 5 years postoperation showing a progression in volarward migration of the proximal ulnar stump



Figure 4: Preoperative photos showing the loss of the finger extension (a) Top view, (b) right side view, and (c) front view. Subsequently, the patient underwent the modified Sauvé–Kapandji procedure and finger extensor tendon reconstruction with a palmaris longus tendon graft. The 5-year postoperative photos demonstrate the restoration of finger extension: (d) top view, (e) left side view, and (f) front view

Table 1: Demographic data of the patients and tendon procedures

Patient	Side	Sex	Age (years)	Extensor tendon rupture	Tendon repair	Follow-up (months)	Larsen grade
1	Right	Female	37	EDC IV	PL tendon graft	47	IV
2–1	Left	Male	78	EDC III–V and EDM	PL tendon graft	62	IV
2–2	Right	Male	79	EDC IV–V and EDM	PL tendon graft	49	V
3–1	Left	Female	37	Nil	Nil	79	IV
3–2	Right	Female	40	EDC III–V and EDM	PL tendon graft	24	V
4	Left	Female	65	EDC V and EDM	End-to-side tendon repair with EDC IV	12	IV
5	Left	Male	73	Nil	Nil	27	III
6–1	Right	Female	49	Nil	Nil	98	III
6–2	Left	Female	57	EDC IV–V and EDM	PL tendon graft	12	III

EDC: Extensor digitorum communis, EDM: Extensor digiti minimi, PL: Palmaris longus

stump. Early postoperative measurements showed substantial narrowing compared to the preoperative ones in the radioulnar width of the wrist. However, this difference was not significant. The Gilula index, which represents ulnar carpal translation, revealed no significant change at the final visit.

The pre- and postoperative clinical outcomes are listed in Table 4. The final postoperative wrist supination was distinguishable from the preoperative stage, indicating significant improvement. However, the final postoperative wrist flexion revealed a significantly decreased range of motion.

Table 2: Radiographic parameters of the ulna stump

Radiographic parameters	Early	Final	P
Ulnar distance (mm)	23.6 (20.9–25.6)	21.5 (18.1–23.0)	0.008*
Ulnar gap (mm)	8.1 (5.7–12.5)	8.8 (6.6–11.0)	0.678
Radioulnar distance (mm)	8.9 (4.6–12.8)	6.3 (3.6–9.7)	0.021*
Dorsopalmar distance (mm)	2.3 (–5.4–5.3)	–2.3 (–8.1–0.84)	0.008*

*Statistical significance. Data presented as median (range)

Table 3: Radiographic parameters related to the ulnar carpal translation

	Preoperative	Final	P
Gilula index	0.31 (0.09–0.59)	0.34 (0.21–0.64)	0.401
Radioulnar width of the wrist (mm)	42.77 (36.14–51.74)	41.32 (34.21–48.85)	0.314

Data presented as median (range)

Table 4: Functional assessment and clinical outcome

	Preoperative	Final	P
Supination (°)	60 (30–85)	90 (60–90)	0.012*
Pronation (°)	70 (40–85)	70 (40–80)	0.052
Flexion (°)	20 (10–50)	10 (10–30)	0.027*
Extension (°)	15 (10–30)	15 (10–80)	0.932
Grip strength (lbs)	9 (3–43)	18 (4–36)	0.066
VAS	5 (3–8)	3 (0–6)	0.068
QuickDASH score	38.6 (31.8–75.0)	15.9 (2.3–45.5)	0.021*
MMWS	45 (10–50)	55 (30–75)	0.028*

*Statistical significance. Data presented as median (range). lbs: Pounds, VAS: Visual Analog Scale, DASH: Disabilities of the Arm, Shoulder, and Hand, MMWS: Modified Mayo wrist score

A significant improvement in the QuickDASH score suggested that the operation had a positive effect on recovery from functional impairment in the upper extremities. The prognosis of wrist function and ability to work or perform daily activities were assessed using the MMWS, indicating the advancement of clinical outcomes. The pain and functional scores of specific patients showed that patients 1 and 4 had worsened pain scores with increased QuickDASH scores, indicating more severe disabilities owing to metacarpophalangeal (MCP) and thumb carpometacarpal joint (CMCJ) osteoarthritis. However, higher and lower MMWS scores were observed in these two patients, respectively. Six wrists with tendon reconstruction regained finger extension and had no re-ruptures; however, two wrists showed an extension lag in the MCP joint. No wound infections or drilled-hole breakages were observed during the follow-up.

In cases with clear postoperative posteroanterior radiographs, DRUJ union was recognized as the connection of the bony cortex between the distal ulnar stump and the distal radius at the DRUJ. In cases with unobvious radiographs, indirect signs were evaluated in order to judge the bony union of DRUJ arthrodesis. There was no screw loosening, backout, or breakage after years of follow-up, and patients did not report prominent pain over the DRUJ region. In our series, all patients achieved bone union of the DRUJ arthrodesis.

Other surgeries were performed on two patients. Patient 1 underwent extensor tendon reconstruction with an extensor indicis proprius tendon transfer because of thumb extensor tendon disruption on the ipsilateral hand 1 year later. The MMWS decreased from 50 to 30. Patient 3 underwent Thompson distraction and interpositional arthroplasty to treat the thumb CMCJ osteoarthritis of the same hand 11 months later. The MMWS improved from 10 to 40. Both patients still had pain at other small joints in the hand but not the DRUJ after the surgeries. There was no extension lag in Patient 1 and the thumb was able to achieve opposition and circumduction in Patient 3.

DISCUSSION

This case series aimed to demonstrate the radiographic and clinical outcomes of patients with advanced-stage rheumatoid wrists after the modified S–K procedure without additional procedures of inserting a resected ulnar segment in the DRUJ.^[12–16,18] Six wrists underwent tendon reconstruction owing to extensor tendon rupture and had no re-ruptures. Although radiographic examinations showed instability of the proximal ulnar stump at the latest follow-up, significant improvements were observed in postoperative wrist supination, QuickDASH score, and MMWS. Two patients subsequently underwent another surgery that was not directly related to the complications or effects of the first procedure. No complications were observed. Nevertheless, patients should be reminded of possible secondary surgeries for other degenerative conditions arising from RA progression.

The most feared complications following the S–K procedure are instability of the proximal ulnar stump and ulnar carpal translation.^[12,13,18] Radioulnar convergence and instability of the proximal ulnar stump causing pain after the S–K procedure have been reported.^[12,13] A modified S–K procedure with stabilization of the ulnar stump decreased proximal ulnar stump pain.^[14,16] Currently, most studies on the modified S–K procedure have revealed a radiologically unstable proximal ulnar stump without evident ulnar stump pain.^[15,16,18,19] Regarding the radioulnar distance, some studies identified radioulnar convergence, indicating radial deviation of the stump on follow-up radiographs.^[15,18] Regarding the dorsopalmar distance, postoperative radiographs showed a negative measurement, representing volarward migration of the proximal ulnar stump in several studies.^[15,16,19] Two studies reported statistically significant postoperative changes in the dorsopalmar distance;^[15,16] however, no significant difference was observed in another study.^[19] In our study, a similar tendency was observed, with a significant difference in radioulnar convergence and volarward migration of the proximal ulnar stump between the early and final stages after the index procedure; however, we observed no proximal ulnar stump impingement. Radiological proximal stump instability may be a concern; however, the QuickDASH score and MMWS showed significant improvements in daily activity. Tomori *et al.* postulated that proximal ulnar stump pain may

not be caused by radial or dorsal deviation of the proximal ulnar stump but by other dynamic factors.^[21]

Besides the instability of the proximal ulnar stump, ulnar carpal translation is another feared complication. Theoretically, ulnar and palmar carpal translations are prevented after the S–K procedure.^[11,22] However, in a recent study that included patients with osteoarthritic and rheumatoid wrists after the modified S–K procedure, increased ulnar carpal translation was observed.^[18] Preservation of the ulnar head after an ulnar leveling procedure can provide better ulnar support to the wrist with a wider radiocarpal shelf and prevent ulnar carpal translation.^[4,10,15] To define the width of the radiocarpal shelf, Ota *et al.* measured the radioulnar width of the wrist and discovered that the loss between the 1-month postoperative and preoperative widths of the wrists was correlated with additional ulnar carpal translation.^[18] A short segment from the ulnar shaft was resected and rotated to bridge the gap in the DRUJ, and the radioulnar width of the wrist was maintained throughout the follow-up.^[18] In our study, the gap in the DRUJ was not bridged with any bony segments. Although narrowing of the radioulnar width of the wrist was observed throughout the follow-up, no significant difference was observed between the early postoperative and preoperative measurements. The latest radiographs demonstrated no deterioration in ulnar carpal translation. We postulate two possible reasons for this finding. First, the radioulnar width of the wrist was preserved by avoiding over-compression, as mentioned in the surgical technique. Second, because the radioulnar width of the wrist was 5 mm proximal to the distal end of the ulnar head, it could be affected by the shortening of the distal ulnar stump and showed a substantial decrease in radioulnar width at the early postoperative stage.

In the pronation-supination arc, increased supination after the S–K procedure or a modification for the stabilization of the proximal ulnar stump has achieved statistical significance in several studies.^[10,15,23,24] Similarly, better supination motion was observed in our study. Using the same procedure used in this study, Minami *et al.* reported improved postoperative wrist extension but not flexion.^[15] In the present study, the patients showed significantly worse range of motion in wrist flexion but not in extension. Serum markers of disease activity may provide a better picture of the correlation between disease progression, range of motion, and functional outcomes.^[4] However, we lacked a comprehensive record of serum markers of disease activity owing to the retrospective nature of the analysis. Regarding functional outcomes, patients with inflammatory arthritis reported an improved QuickDASH score 12 months after the modified S–K procedure with partial ECU or pronator quadratus tenodesis.^[24] In patients with rheumatoid wrists who underwent the modified S–K procedure with the proximal ulnar stump covered with periosteum and soft tissue, the QuickDASH score improved significantly.^[23] The postoperative improvement in the QuickDASH score

was also statistically significant in our case series, in which the modified S–K procedure was performed with ECU tenodesis. However, two additional operations for disorders caused by disease progression were performed, and pain deterioration because of new-onset pain in the CMCJ and MCP joints was observed in the latest follow-up visit. As pain is one of the main influencers of validated functional scores, the increased QuickDASH scores in both cases correlated with this condition. We also recorded the MMWS during the postoperative follow-up, which has not been broadly reported in other studies. The significant difference between the preoperative and postoperative MMWS implied a good prognosis for wrist function and return to working or performing daily activities. In contrast to the limited evidence of functional outcomes after the index procedure and the vagueness of the severity of preoperative rheumatoid wrists in previous studies, our research provides a point of view regarding changes in wrist range of motion, performance enhancement, and the possibility of disease progression in other joints in patients with advanced-stage rheumatoid wrists postsurgery.

This case series had some limitations. First, this was a retrospective study with a small sample size, which may be explained by the rarity of patients with advanced-stage rheumatoid wrists. Hence, the small sample size limited the significance of this study. In addition, the retrospective design could have introduced a selection bias. A selection bias in retrospective studies can lead to better outcomes. Second, medium-term follow-up may not provide the whole picture of disease progression and final outcomes. Finally, this study lacked direct comparisons with other treatment options. Further studies are warranted to elaborate on the details of the disease, such as the clinical relevance of radiological parameters and the superiority of the modified S–K procedure.

CONCLUSIONS

This study supports the efficacy of the modified S–K procedure without inserting a resected ulnar segment between the DRUJ in treating patients with advanced-stage rheumatoid wrists. However, patients must be informed that other disorders due to RA progression may occur, and a secondary procedure may be required. The operation prevented the narrowing of the wrist width and ulnar carpal translation, but did not prevent radiologic instability of the ulnar stump, as indicated by radioulnar convergence and volarward migration. Nevertheless, radiological instability does not always result in suboptimal results. The clinical relevance of radioulnar convergence and volarward migration of the proximal ulnar stump should be assessed in further comparative studies.

Author contributions

Conceptualization, C.-H. Wu; Methodology, C.-H. Wu; Formal Analysis, K.-Y. Liu; Investigation, K.-Y. Liu; Resources, C.-H. Wu; Data Curation, K.-Y. Liu, C.-H. Wu; Writing – Original

Draft Preparation, K.-Y. Liu; Writing – Review & Editing – C.-H. Wu; Supervision – C.-H. Wu, C.-H. Ma, I.-M. Jou, Y.-K. Tu; Project Administration, C.-H. Wu. All authors have read and agreed to the final version of the manuscript.

Data availability statement

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

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Conflicts of interest

Dr. Chin-Hsien Wu, an editorial board member at *Formosan Journal of Musculoskeletal Disorders*, had no role in the peer review process of or decision to publish this article. All authors declared no conflicts of interest in writing this paper.

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